# BMC Helix Technical Bootcamp - ITSM/ITOM Container

## Lab 6: Performing the BMC Helix Service Management Installation

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# Lab 6: Performing the BMC Helix Service Management Installation

**Objective:** In this lab you will install BMC Helix Service Management platform and applications by using the **HELIX_ONPREM_DEPLOYMENT** Jenkins pipeline.

The **HELIX_ONPREM_DEPLOYMENT** Jenkins pipeline runs various deployment pipelines, in a sequence, and configures the BMC Helix Innovation Suite platform and service management applications.

But prior to performing the BMC Helix Service Management installation, you must complete the following actions to prepare the environment for installation.

- Verify Domain Name System (DNS) for applications
- Create a BMC Helix Single Sign-On realm

Refer to the full product documentation for complete implementation details. The steps provided in this lab document are simplified and tailored to our specific environment. It is recommended that you review the full documentation before beginning any lab work:

https://docs.bmc.com/xwiki/bin/view/Service-Management/On-Premises-Deployment/BMC-Helix-Service-Management-Deployment/brid25101/Installing/Preparing-for-installation/Setting-up-the-installation-environment/

## Task 1: Prepare for Installation-Verify DNS for applications

In this task you will make sure that you configure DNS for the BMC Helix Service Management applications so that you can access the applications by using a specific URL format.

Example: **Mid Tier URL:** <*CUSTOMER_SERVICE*>-<*ENVIRONMENT*>.<*CLUSTER_DOMAIN*>

In this training environment CUSTOMER_SERVICE=**is**, ENVIRONMENT=**poc**, and CLUSTER_DOMAIN=**training.local**, so the Mid-Tier URL will be **is-poc.training.local**. These values will be defined in the appropriate parameters in the Jenkins HELIX_ONPREM_DEPLOYMENT deployment pipeline configured below.

1. Using **MobaXTerm**, log into the **Deployment Engine** as **root/password**.
2. Run the **nslookup** command from the deployment engine and confirm these FQDNs resolve to the external load balancer IP 192.168.1.20 in this lab environment:

```
nslookup is-poc.training.local
```

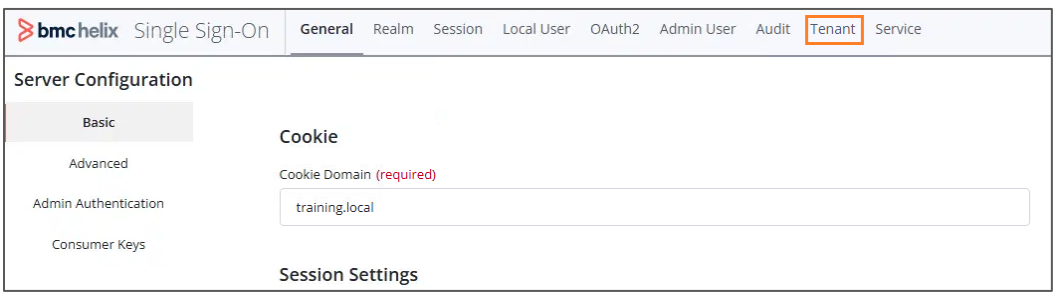
```
Last login: Thu Oct 30 15:14:48 2025 from 192.168.1.11
[root@deploymentengine ~]# nslookup is-poc.training.local
Server:         192.168.1.11
Address:        192.168.1.11#53

Name:   is-poc.training.local
Address: 192.168.1.20

[root@deploymentengine ~]#
```

## Task 2: Prepare for Installation-Configure BMC Helix Single Sign-On

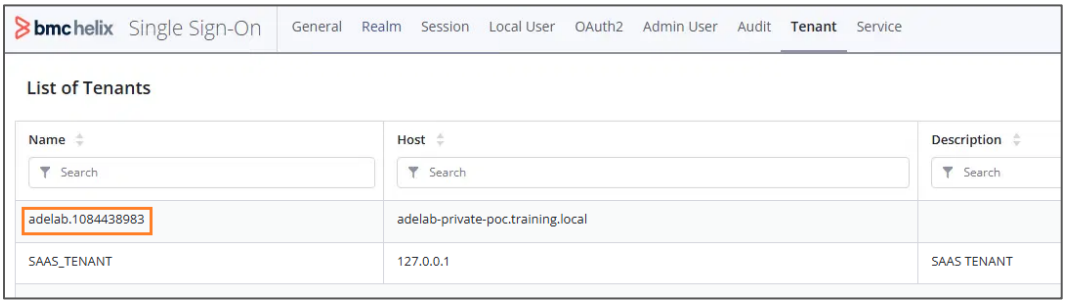
BMC Helix Single Sign-On was installed during the BMC Helix Platform services deployment. In this task you will configure BMC Helix Single Sign-On.

1. In **Chrome**, log into BMC Helix Single Sign-on:
   - **https://adelab.training.local/rsso**
   - Username: **Admin**
   - Password: **RSSO#Admin#**

2. On the menu, click **Tenant**.




3. In the **List of Tenants**, locate the tenant with a name in the following format: **adelab.XXXXXXXX**




4. Copy the value from the **Name** column.
5. Paste the value into Notepad and save it for reference - you will need it two more times later during this lab.
6. In the **List of Tenants**, locate the tenant with a name **SAAS_TENANT**.
7. Under the **Action** column, click to select the **SAAS_TENANT**. "**Tenant SAAS_TENANT is selected**" confirmation message is displayed on the screen.

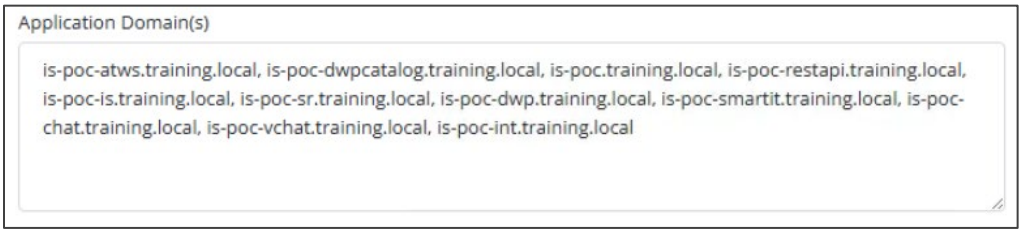
8. On the toolbar, click **Realm**.
9. Click **Add Realm**.
10. In the **General** tab, enter the following details:

| Field | Value |
|---|---|
| **Realm ID** | is-poc<br><br>**Note**: The Realm ID is a combination of <CUSTOMER_SERVICE>-<ENVIRONMENT>. In our training environment, CUSTOMER_SERVICE=is, ENVIRONMENT=poc, so the field value is **is-poc.** |
| **Application Domain(s)** | is-poc-atws.training.local, is-poc-dwpcatalog.training.local, is-poc.training.local, is-poc-restapi.training.local, is-poc-is.training.local, is-poc-sr.training.local, is-poc-dwp.training.local, is-poc-smartit.training.local, is-poc-chat.training.local, is-poc-vchat.training.local, is-poc-int.training.local |
| **Tenant** | Paste in the Tenant Name that you copied in **Step 4** with the format: **adelab.XXXXXXXX**<br><br>**Note:** Ensure there are NO trailing white spaces after tenant name as this can cause installation issues. |

11. Click the **Authentication** tab and enter the following details:

| Field | Value |
|---|---|
| **Authentication Type** | AR |
| **Host** | platform-user-ext.helixis |
| **Port** | 46262 |

12. Click **Add**.
13. From the **Realm** tab, click the **pencil icon** to open and verify the Application Domain(s) for the **is-poc** Realm ID.
14. Verify the **Application Domain(s) field** contains the correct information copied earlier in the lab.




**VERY IMPORTANT:** Incorrect values can be copied when using the copy and paste feature to update this field. Incorrect information in this field will result in errors to be found by the

HITT script in Task 6. Check the list of domains carefully to ensure there are no illegal or missing characters in the list.

## Task 3: Download and Copy cacerts file

In this task, you will download a default cacerts file from docs.bmc.com and run a command to import the provided certificate into it, so the application pods can trust the self-signed certificate used in the lab.

Refer to the full product documentation for complete implementation details. The steps provided in this lab document are simplified and tailored to our specific environment. It is recommended that you review the full documentation before beginning any lab work:

https://docs.bmc.com/xwiki/bin/view/Service-Management/On-Premises-Deployment/BMC-Helix-Service-Management-Deployment/brid25101/Installing/Preparing-for-installation/Preparing-to-use-self-signed-or-custom-CA-certificates/

1. As **root** create a **cert** directory in the root directory:

```
mkdir /root/cert
```

2. Open this page in the BMC Documentation website to download the **cacerts** file.

https://docs.bmc.com/xwiki/bin/view/Service-Management/On-Premises-Deployment/BMC-Helix-Service-Management-Deployment/brid23304/Installing/Preparing-for-installation/Preparing-to-use-self-signed-or-custom-CA-certificates

3. Locate the section **To create a self-signed or custom CA certificate** section.

> To create a self-signed or custom CA certificate
>
> 1. Download the cacerts file.
> Do not use the **custom_cacerts.pem** or **cacerts** file located in the helix-on-prem-deployment-manager/commons/certs folder that you used for BMC Helix Platform Common Services installation.

**Note:** You will need to login to the documentation website

4. On the **PDFs** page, click the **cacerts** file to download.

5. Copy the **cacerts file** from the Windows System to **/root/cert** using MobaXterm or WinSCP.
6. Import custom certificate to out of box ITSM cacerts using this command and type "yes" to trust the certificate:

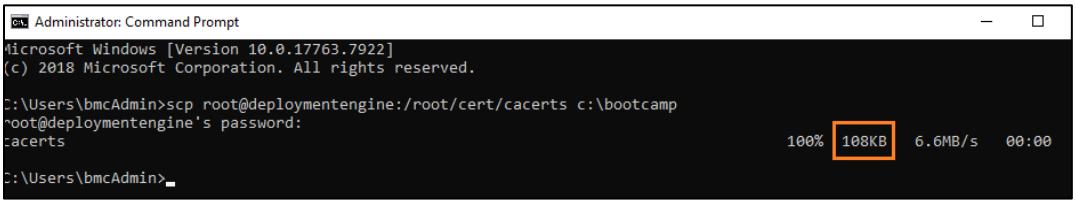
```
keytool -importcert -v -alias bootcamp -file
/root/class_kit/certs/custom_cacert.pem -keystore
/root/cert/cacerts -storepass changeit
```

7. Using **File Explorer**, create a **bootcamp** directory on the **C drive**.

8. Using **Windows command prompt**, copy the **cacerts** file to **c:\bootcamp** with this command:

```
scp root@deploymentengine:/root/cert/cacerts c:\bootcamp
```

9. Verify the copied cert is approx. 108KB. A corrupt cert will cause the installation to fail.




Note: The **cacerts** file must be copied to the landing desktop so it can be uploaded to the **HELIX_ONPREM_DEPLOYMENT** pipeline while performing the installation.

## Task 4: Configure the HELIX_ONPREM_DEPLOYMENT pipeline

In this task you will configure the HELIX_ONPREM_DEPLOYMENT pipeline with all the required values. In the PRODUCTS DEPLOY section you will only select HELIX_GENERATE_CONFIG. This will save the values and then you will run the HITT script (that was downloaded in the previous lab) to verify that the values supplied are correct and an installation should succeed.

Refer to the full product documentation for complete implementation details. The steps provided in this lab document are simplified and tailored to our specific environment. It is recommended that you review the full documentation before beginning any lab work:

https://docs.bmc.com/xwiki/bin/view/Service-Management/On-Premises-Deployment/BMC-Helix-Service-Management-Deployment/brid25101/Installing/Performing-the-BMC-Helix-Service-Management-installation/

1. In **Chrome**, log into the **Jenkins Dashboard:**
   - **http://deploymentengine:8080**
   - Username: **Admin**
   - Password: **Passw0rd**!

   **Note**: You may get a **ssl** error, if you do type the following into the URL **deploymentengine:8080** this should take you to the login screen, alternatively, clear the browser history.

2. Select the **HELIX_ONPREM_DEPLOYMENT** pipeline and click **Build with Parameters**.

   **Note:** The **Build with Parameters** option is used to start a new build of the job and manually set or change the parameter values before running it.

3. In the **INFRASTRUCTURE** section set or verify the following values:

| Parameter | Value |
|---|---|
| HELM_BINARY | helm |
| CUSTOM_BINARY_PATH | Unchecked |
| AGENT | git-deploymentengine.training.local |
| CHECKOUT_USING_USER | github |
| KUBECONFIG_CREDENTIAL | kubeconfig |

4. In the **CODE** section set or verify the following values:

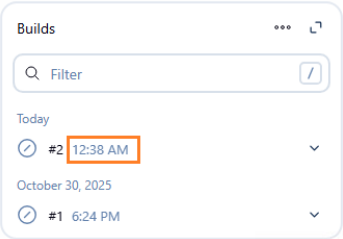
| Parameter | Value |
|---|---|
| GIT_USER_HOME_DIR | /home/git |
| GIT_REPO_DIR | ssh://git@deploymentengine/home/git/git_repo |

5. In the **CUSTOMER-INFO** section set or verify the following values:

| Parameter | Value |
|---|---|
| IS_CLOUD | Unchecked |
| ROUTE_ENABLED | Unchecked |
| ROUTE_TLS_ENABLED | Unchecked |
| OS_RESTRICTED_SCC | Unchecked. We are not using OpenShift. |
| DEPLOYMENT_MODE | FRESH |
| CLUSTER | kubernetes-admin@kubernetes |

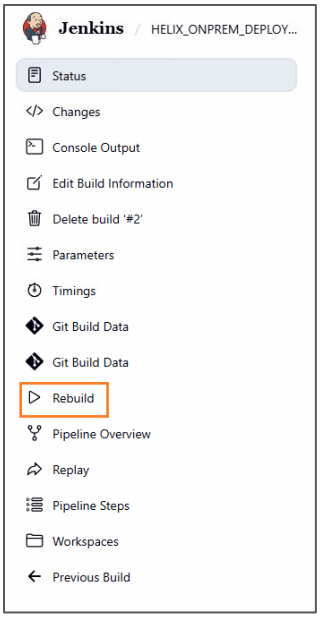
| | |
|---|---|
| | **NOTE:** The following commands can be used to get above value:<br>1. ssh deployment engine<br>2. cat /root/.kube/config \| grep current-context<br><br>[root@deploymentengine ~]# cat /root/.kube/config \| grep current-context<br>current-context: kubernetes-admin@kubernetes<br>[root@deploymentengine ~]# |
| **CUSTOMER_NAME** | ispoc |
| **IS_NAMESPACE** | helixis |
| **CUSTOMER_SERVICE** | is |
| **ENVIRONMENT** | poc |
| **INGRESS_CLASS** | nginx |
| **CLUSTER_DOMAIN** | training.local |
| **INPUT_CONFIG_METHOD** | Generate_Input_File |
| **INPUT_CONFIG_FILE** | *Not in use* |
| **CACERTS_FILE** | Upload cacert from c:\bootcamp |
| **CACERTS_SSL_TRUSTSTORE_PASSWORD** | *leave defaults* |
| **DB_SSL_CERT** | Not in use |
| **CUSTOMER_SIZE** | C |
| **HELM_NODE** | deploymentengine.training.local |

6. Click **Build**. Since not all parameters have been entered, this run is expected to abort.
7. In the **Builds** section, click on the most recent run.




8. On the left-hand menu, click **Rebuild** to continue entering in additional parameters.

   **IMPORTANT**: The **Rebuild** option allows you to rerun the pipeline using the same saved parameters. Because Jenkins has a 30-minute timeout and this pipeline requires multiple parameters, it is recommended to periodically save your progress by clicking **Rebuild** and then aborting the current run.
   Do **not** use **Build with Parameters**, as this will start a new run without any of your previously saved parameter values.




9. In the **PRODUCTS** section set or verify the following values:

| Parameter | Value |
|---|---|
| HELIX_VIRTUALCHAT | Checked |
| HELIX_OPENFIRE | Checked |
| HELIX_DWP | Checked |
| HELIX_DWPA | Checked |
| HELIX_CLOUD_ACTIONS | **Unchecked** |
| HELIX_BWF | Checked |
| HELIX_MCSM | Checked |
| HELIX_ITSM_INSIGHTS | **Unchecked** |
| HELIX_TSOMPLUGIN | **Unchecked** |
| HELIX_SMARTAPPS_CSM | **Unchecked** |
| HELIX_SMARTAPPS_FAS | **Unchecked** |
| HELIX_DRIFT_MANAGEMENTPLUGIN | **Unchecked** |
| HELIX_CLAMAV | **Unchecked** |
| HELIX_NETOPS | **Unchecked** |
| HELIX_GPT | **Unchecked** |

10. In the **LOGGING CONFIGURATION** section set or verify the following values:

| Parameter | Value |
|---|---|
| SIDECAR_SUPPORT_ASSISTANT_FPACK | Checked |
| SUPPORT_ASSISTANT_CREATE_ROLE | Checked |
| SUPPORT_ASSISTANT_SERVICE_ACCOUNT | default |
| SIDECAR_FLUENTBIT | **Unchecked** |
| SIDECAR_FLUENT_DETAIL_LOG | **Unchecked** |

11. In the **PRODUCT-DEPLOY** section set or verify the following values. You will come back and select more products in this section after you have tested the pipeline with the Helix IS Triage Tool (HITT):

| Parameter | Value |
|---|---|
| **HELIX_GENERATE_CONFIG** | Checked |
| **HELIX_PLATFORM_DEPLOY** | **Unchecked** |
| **HELIX_NONPLATFORM_DEPLOY** | **Unchecked** |
| **HELIX_CONFIGURE_ITSM** | **Unchecked** |
| **HELIX_SMARTAPPS_DEPLOY** | **Unchecked** |
| **SUPPORT_ASSISTANT_TOOL** | **Unchecked** |
| **HELIX_INTEROPS_DEPLOY** | **Unchecked** |
| **HELIX_FULL_STACK_UPGRADE** | **Unchecked** |
| **HELIX_POST_DEPLOY_CONFIG** | **Unchecked** |
| **HELIX_DR** | **Unchecked** |
| **SCALE_DOWN** | **Unchecked** |
| **HELIX_RESTART** | **Unchecked** |

12. In the **DATABASE_REGISTRY_STORAGE_PASSWORDS_SECRETS_DETAILS** section, set or verify the following values:

| Parameter | Value |
|---|---|
| **REGISTRY_TYPE** | DTR |
| **HARBOR_REGISTRY_HOST** | deploymentengine.training.local |
| **IMAGE_REGISTRY_USERNAME** | dev-itom-class |
| **IMAGE_REGISTRY_PASSWORD** | Admin12345 |
| **DB_TYPE** | postgres |
| **DB_SSL_ENABLED** | **Unchecked** |
| **DB_JDBC_URL** | *leave blank* |
| **DB_PORT** | 5432 |
| **ORACLE_SERVICE_NAME** | *leave blank* |
| **DATABASE_HOST_NAME** | postgres-svc.postgres |
| **DATABASE_ADMIN_USER** | postgres |
| **DATABASE_ADMIN_ PASSWORD** | bmcAdm1n |
| **PLATFORM_SR_DB_JDBC_URL** | *leave blank* |
| **DATABASE_RESTORE** | Checked to create required AR database tables |
| **AR_DB_CASE_SENSITIVE** | **Unchecked** |
| **IS_DATABASE_ALWAYS_ON** | **Unchecked** |
| **LOGS_ELASTICSEARCH_ HOSTNAME** | *leave blank* |
| **LOGS_ELASTICSEARCH_TLS** | Checked |
| **LOGS_ELASTICSEARCH_PASSWORD** | *leave defaults* |

| | |
|---|---|
| **FTS_ELASTICSEARCH_HOSTNAME** | opensearch-logs-data.helixade |
| **FTS_ELASTICSEARCH_PORT** | 9200 |
| **FTS_ELASTICSEARCH_USERNAME** | bmcuser |
| **FTS_ELASTICSEARCH_USER_PASSWORD** | Es_L0g#p@SS<br><br>**Note:** This information can be obtained from secrets.bak file. |
| **FTS_ELASTICSEARCH_SECURE** | Checked |
| **AR_LOCALE_TO_INSTALL** | *leave blank* |
| **IMAGESECRET_NAME** | isharbor-secret |
| **BAKEDUSER_HANNAH_ADMIN_PASSWORD** | Passw0rd!<br><br>**NOTE:** hannah_admin user in Helix Platform Services is different from hannah_admin user in Innovation Suite |
| **AR_DB_NAME** | is_db |
| **AR_DB_USER** | ARAdmin |
| **AR_DB_PASSWORD** | AR#Admin# |
| **PLATFORM_SR_DB_USER** | *leave blank* |
| **PLATFORM_SR_DB_PASSWORD** | *leave defaults* |
| **AR_SERVER_APP_SERVICE_PASSWORD** | AR#Admin# |
| **AR_SERVER_DSO_USER_PASSWORD** | AR#Admin# |
| **AR_SERVER_MIDTIER_SERVICE_PASSWORD** | AR#Admin# |
| **VC_RKM_USER_NAME** | vc_admin |
| **VC_RKM_PASSWORD** | AR#Admin# |
| **VC_PROXY_USER_LOGIN_NAME** | vcp_admin |
| **VC_PROXY_USER_PASSWORD** | AR#Admin# |
| **DWP_CONFIG_PRIMARY_ORG_NAME** | dwporg |
| **AR_SERVER_ALIAS** | onbmc-s |
| **PLATFORM_ADMIN_PLATFORM_EXTERNAL_IPS** | [192.168.1.21]<br><br>**Note:** Value must include brackets [ ]. Use IP address of any worker node:<br>• $ ssh deploymentengine - root/password<br>• $ kubectl get nodes -o wide |
| **ENABLE_PLATFORM_INT_NORMALIZATION** | **Unchecked** |
| **MIDTIERCACHEBUILDER_TRIGGER_PRELOAD** | **Unchecked** |
| **MIDTIERCACHEBUILDER_SCHEDULE** | *leave defaults* |

| | |
|---|---|
| **BWF_DEPLOY_SAMPLE_CONTENT_PACK** | **Unchecked** |
| **DWP_DEPLOY_SAMPLE_CONTENT_PACK** | **Unchecked** |
| **CLOUDACTIONS_DEPLOY_SAMPLE_CONTENT_PACK** | **Unchecked** |
| **AR_DATETIME** | *leave blank* |
| **AR_LOCALE** | *leave blank* |

13. In the **RSSO_PARAMETERS** section, set or verify the following parameters:

| Parameter | Value |
|---|---|
| **RSSO_URL** | https://adelab.training.local/rsso<br><br>**Note:** Do NOT add an extra '/' to the end of the path |
| **RSSO_ADMIN_USER** | Admin |
| **RSSO_ADMIN_PASSWORD** | RSSO#Admin# |
| **TENANT_DOMAIN** | Copy the tenant name value you gathered in Task 2. This value should be in Notepad with the format: **adelab.XXXXXXXX** |

14. In the **ITSM_INTEROPS_PARAMETERS** section, set or verify the following parameters:

| Parameter | Value |
|---|---|
| **HELIX_PLATFORM_DOMAIN** | training.local |
| **HELIX_PLATFORM_NAMESPACE** | helixade |
| **HELIX_PLATFORM_CUSTOMER_NAME** | adelab |

15. In the **Select the services for interoperability configuration** section, select the services that you want to enable for the tenant that was created during the BMC Helix Platform installation:

| Parameter | Value |
|---|---|
| **BMC_HELIX_ITSM_INSIGHTS** | **Unchecked** |
| **BMC_HELIX_SMART_IT** | Checked |
| **BMC_HELIX_BWF** | Checked |
| **BMC_HELIX_DWP** | Checked |
| **BMC_HELIX_INNOVATION_STUDIO** | Checked |
| **BMC_HELIX_DWPA** | Checked |

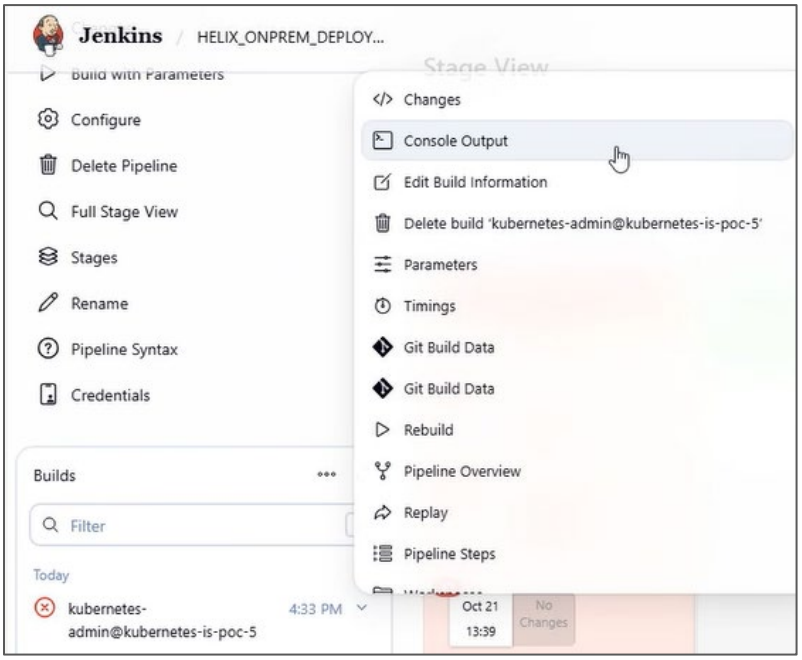
16. Verify that all the above settings are correct.

17. In the **CUSTOMER-INFO** section, verify the **CACERTS_FILE** has been added. This is the only setting that is **NOT** preserved when you rebuild a pipeline run, so you must set it every time.

18. Click **Build (or Rebuild)**. This pipeline run is expected to fail, but you will need to verify exactly how it fails in the next steps.

19. Verify that the pipeline run fails after 2 minutes:

20. When the run fails, view the **Console Output** of the **HELIX_ONPREM_DEPLOYMENT** pipeline. The first run will fail as there is a requirement to approve a signature.



21. Verify you see the following lines in the Console Output:
    - Build HELIX_GENERATE_CONFIG completed: SUCCESS
    - Administrators can decide whether to approve or reject this signature.
    - Finished: FAILURE

```
Build HELIX_GENERATE_CONFIG kubernetes-admin@kubernetes-is-poc-2 completed: SUCCESS
Scripts not permitted to use method org.jenkinsci.plugins.workflow.support.steps.build.RunWrapper getRawBuild. Administrators can decide whether to approve or reject this
signature.
[Pipeline] }
[Pipeline] // stage
[Pipeline] sh
+ rm -rf /home/git/jenkins_node_root_dir/workspace/HELIX_ONPREM_DEPLOYMENT/environments/customer/customCerts
++ basename ssh://git@deploymentengine/home/git/git_repo/CUSTOMER_CONFIGS/onprem-remedyserver-config.git
++ cut -d. -f 1
+ rm -rf /home/git/jenkins_node_root_dir/workspace/HELIX_ONPREM_DEPLOYMENT/environments/onprem-remedyserver-config
++ basename ssh://git@deploymentengine/home/git/git_repo/CUSTOMER_CONFIGS/onprem-remedyserver-config.git
++ cut -d. -f 1
+ rm -rf /home/git/jenkins_node_root_dir/workspace/HELIX_ONPREM_DEPLOYMENT/onprem-remedyserver-config
+ rm -rf /tmp/InputFile
[Pipeline] echo
Exception occured : org.jenkinsci.plugins.scriptsecurity.sandbox.RejectedAccessException: Scripts not permitted to use method
org.jenkinsci.plugins.workflow.support.steps.build.RunWrapper getRawBuild
[Pipeline] sh
+ rm -rf '/home/git/jenkins_node_root_dir/workspace/HELIX_ONPREM_DEPLOYMENT/environments/customer/*'
++ basename ssh://git@deploymentengine/home/git/git_repo/CUSTOMER_CONFIGS/onprem-remedyserver-config.git
++ cut -d. -f 1
+ rm -rf /home/git/jenkins_node_root_dir/workspace/HELIX_ONPREM_DEPLOYMENT/environments/onprem-remedyserver-config
++ basename ssh://git@deploymentengine/home/git/git_repo/CUSTOMER_CONFIGS/onprem-remedyserver-config.git
++ cut -d. -f 1
+ rm -rf /home/git/jenkins_node_root_dir/workspace/HELIX_ONPREM_DEPLOYMENT/onprem-remedyserver-config
+ rm -rf /tmp/InputFile
[Pipeline] }
[Pipeline] // node
[Pipeline] }
[Pipeline] // timeout
[Pipeline] End of Pipeline
Finished: FAILURE
```

**IMPORTANT**: If the pipeline fails for any reason other than what is shown in the screenshot above, one or more of the entered parameters may be incorrect. Review the console output to identify and resolve the issue and remember to reattach the cacerts.file before re-running the pipeline.

## Task 5: Address Failed Pipeline Execution–Approve Signature

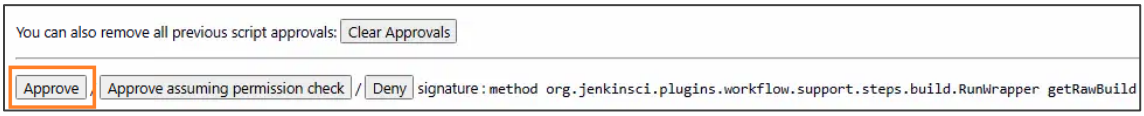
In this task you will approve the two scripts that are currently not approved.

1. In the **Console Output**, click **Administrators can decide whether to approve or reject this signature**.

```
-is-poc-3 completed: SUCCESS
ns.workflow.support.steps.build.RunWrapper getRawBuild. Administrators can decide whether to approve or reject this signature.
```

2. You are redirected to the **Jenkins Script Approval** page. Click **Approve** to view the pending script approval.

3. Click **Approve** to approve the first signature.



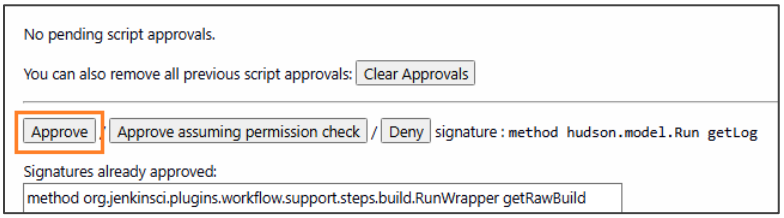

4. Return to the **Jenkins Dashboard** and open the **HELIX_ONPREM_DEPLOYMENT** pipeline.

   - Open the latest pipeline run and click **Rebuild.**
   - Re-add the **cacerts file**.
   - Click **Rebuild**.

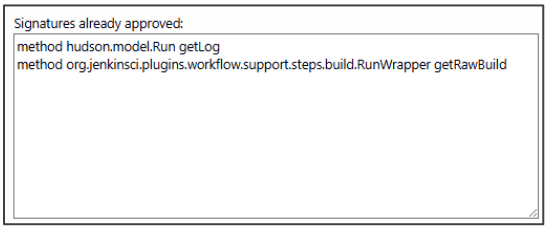
   **NOTE:** the certificate must be re-added each time the pipeline is rebuilt.

5. Again the pipeline will fail. When the run fails, with the **Console Output**, click **Administrators can decide whether to approve or reject this signature**.

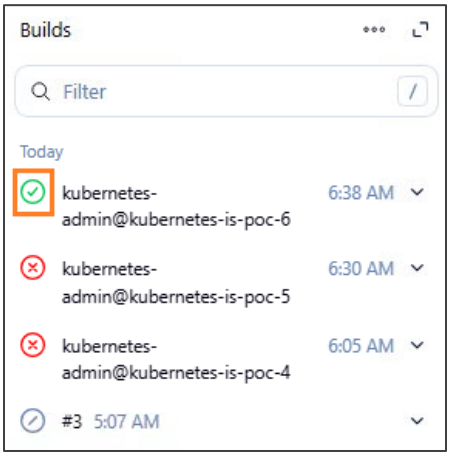
6. Click **Approve** to approve the second signature and **Rebuild** again.




7. Verify 2 signatures are approved.




8. Rebuild the **HELIX_ONPREM_DEPLOYMENT** pipeline and verify that the pipeline runs successfully.



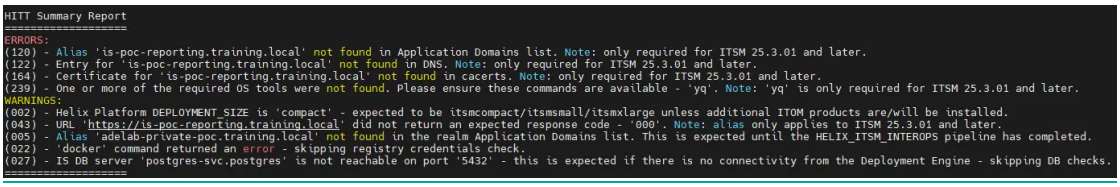

## Task 6: Execute HITT Script

In this task you will execute the hitt script against the last build (successful or not) of the HELIX_ONPREM_DEPLOYMENT pipeline to test for any common configuration problems.

1. In **MobaXterm** log in as **git/password**.
2. Go to **/home/git/hitt** and run the script in **pre-is** mode to validate the values entered in the pipeline fields using the following command:

```
./hitt.sh -m pre-is
```

3. View the **HITT Summary Report**. There are expected errors and warnings that can be ignored.




4. Verify you see errors that end with **Note: only required for ITSM 25.3.01 and later**. These errors can be ignored because it applies to a different version - you are installing 25.1.01.

**IMPORTANT:** If HITT returns any **additiona**l errors, review and resolve them before proceeding. The **hittmsgs.log** file can help identify issues in the pipeline configuration. **Do not continue to Task 7** until all errors have been successfully remediated - proceeding without resolving them will result in installation failure.

## Task 7: Execute Innovation Suite Installation

In this task you will install BMC Helix Service Management platform and applications by using the **HELIX_ONPREM_DEPLOYMENT** Jenkins pipeline. The **HELIX_ONPREM_DEPLOYMENT** Jenkins pipeline runs various deployment pipelines, in a sequence, and configures the BMC Helix Innovation Suite platform and service management applications. BMC Helix Innovation Suite platform and applications will be installed by enabling additional pipelines.

1. In **Chrome**, log into the **Jenkins Dashboard:**
   - **http://deploymentengine:8080**
   - Username: **Admin**
   - Password: **Passw0rd**!

2. From the **HELIX_ONPREM_DEPLOYMENT** pipeline:
   - Open the latest pipeline run and click **Rebuild.**
   - Re-add the **cacerts file**.

   **Note:** The certificate must be re-added each time the pipeline is rebuilt.

3. Go to the **PRODUCT-DEPLOY** section and add the following pipelines:
   - **HELIX_GENERATE_CONFIG**
   - **HELIX_PLATFORM_DEPLOY**
   - **HELIX_NON_PLATFORM_DEPLOY**
   - **HELIX_CONFIGURE_ITSM**
   - **HELIX_SMARTAPPS_DEPLOY**
   - **SUPPORT_ASSISTANT_TOOL**
   - **HELIX_INTEROPS_DEPLOY**
   - **HELIX_POST_DEPLOY_CONFIG**

4. Verify all parameters are updated as required and scroll to the bottom.

5. Click **Rebuild**.

   **IMPORTANT:** The Innovation Suite Installation will run for ~6 hours.

6. To monitor progress, you can check the console output of EACH pipeline. The HELIX_ONPREM_DEPLOYMENT triggers subsequent pipelines so make sure to monitor the pipeline which is running.

7. A successful deployment will display **green** to indicate completion. If any errors occur during the build, use the **HITT script** to troubleshoot and resolve them before proceeding, or refer to the **Rebuilding Pipelines and Troubleshooting** section below. If the build completes without errors, continue to the next task.

   The example below illustrates a successful run:

| Copy Customer Certs | Generate Input Configuration File | Keep Input File inline with Template | Validate RSSO | Helix Remedy Platform | Helix Remedy Apps | Helix Configure ITSM Apps | Helix Deploy Smart Apps | Helix Deploy Support Assistant Tool | Helix ITSM Interops Pipeline | Helix Full Stack Upgrade | Helix Post Deploy Config |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1s | 30s | 1s | 1s | 36min 18s | 15min 35s | 9s | 1h 32min | 59s | 6min 55s | 115ms | 9s |
| 1s | 18s | 1s | 1s | 1h 12min | 31min 11s | 18s | 3h 4min | 1min 58s | 13min 50s | 121ms | 19s |

## *REBUILDING PIPELINES AND TROUBLESHOOTING*

- If HELIX_ONPREM_DEPLOYMENT fails, uncheck previous pipelines that are already run successfully and check only pipelines which need to be run before rebuilding HELIX_ONPREM_DEPLOYMENT
    - For Example: HELIX_ONPREM_DEPLOYMENT is run after HELIX_PLATFORM_DEPLOY completes successfully, the HELIX_ONPREM_DEPLOYMENT fails due to HELIX_NONPLATFORM_DEPLOY pipeline failure. Fix any errors causing HELIX_NONPLATFORM_DEPLOY to fail, uncheck HELIX_PLATFORM_DEPLOY in PRODUCT-DEPLOY section to prevent it from running again, check HELIX_NONPLATFORM_DEPLOY and any other pipelines that need to be run next and rebuild HELIX_ONPREM_DEPLOYMENT

- If HELIX_PLATFORM_DEPLOY needs to be rebuilt due to failure, please note the DATABASE_RESTORE parameter needs to be unchecked if it had completed successfully in the failed run to prevent AR database exists errors Or AR database needs to be deleted and DATABASE_RESTORE can be checked to recreate AR database.

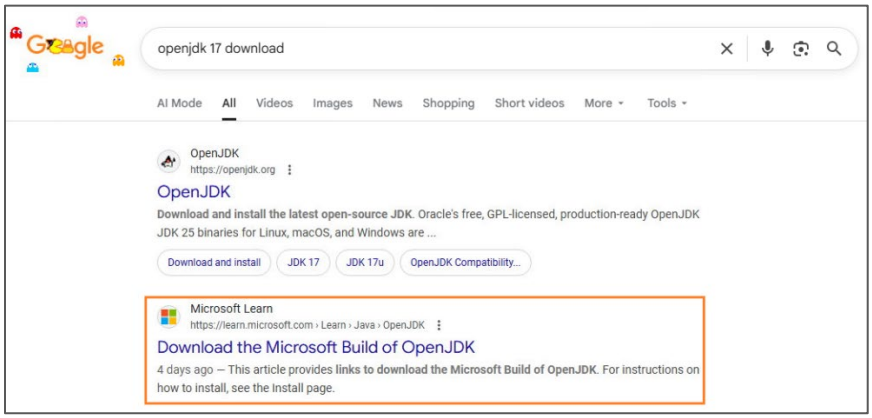
For additional troubleshooting steps and detailed guidance, refer to the full documentation available here:

https://docs.bmc.com/xwiki/bin/view/Service-Management/On-Premises-Deployment/BMC-Helix-Service-Management-Deployment/brid25101/Troubleshooting/Troubleshooting-the-installation-errors/

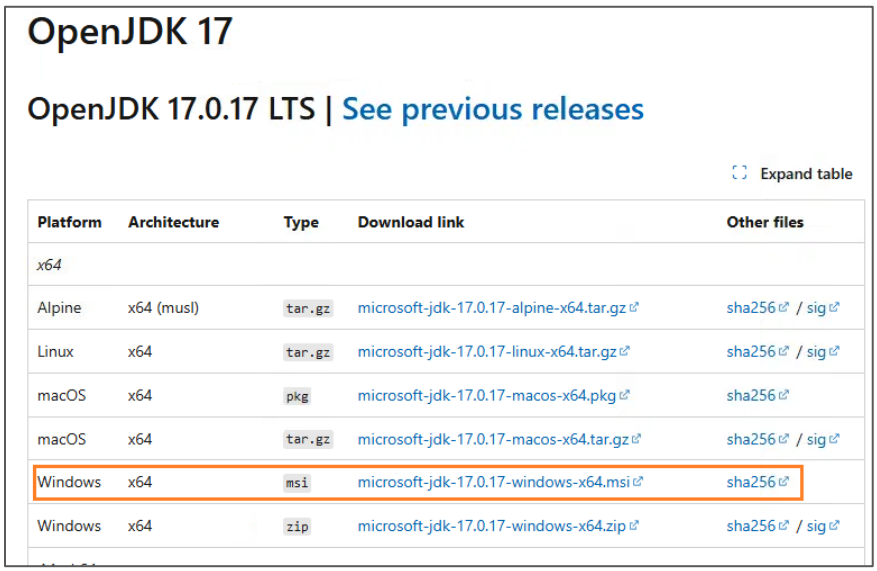
## Task 8: Update Java

In this task you will download and install open jdk version 17 to support the installation of the AR System clients in the next task.

1. Open **Chrome** browser.
2. Using **Google Search** engine, search for **openjdk 17 download.**
3. From the results, select the **Microsoft Learn** result.




4. Download the **microsoft-jdk-17.0.17-windows-x64.msi** installation file.



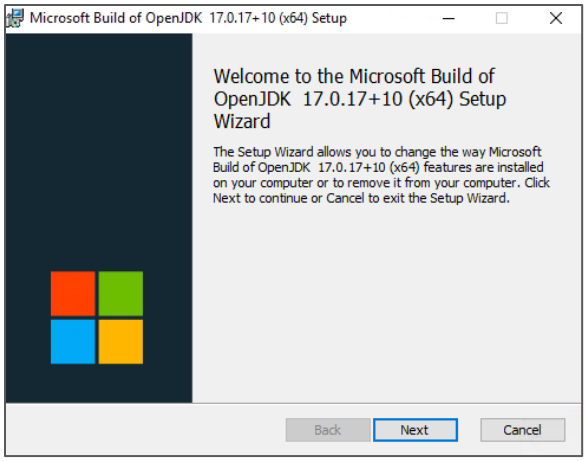
OpenJDK 17

OpenJDK 17.0.17 LTS | See previous releases

Expand table

| Platform | Architecture | Type | Download link | Other files |
|---|---|---|---|---|
| *x64* | | | | |
| Alpine | x64 (musl) | tar.gz | microsoft-jdk-17.0.17-alpine-x64.tar.gz | sha256 / sig |
| Linux | x64 | tar.gz | microsoft-jdk-17.0.17-linux-x64.tar.gz | sha256 / sig |
| macOS | x64 | pkg | microsoft-jdk-17.0.17-macos-x64.pkg | sha256 |
| macOS | x64 | tar.gz | microsoft-jdk-17.0.17-macos-x64.tar.gz | sha256 / sig |
| Windows | x64 | msi | microsoft-jdk-17.0.17-windows-x64.msi | sha256 |
| Windows | x64 | zip | microsoft-jdk-17.0.17-windows-x64.zip | sha256 / sig |

5. Run the installation **accepting all defaults**.

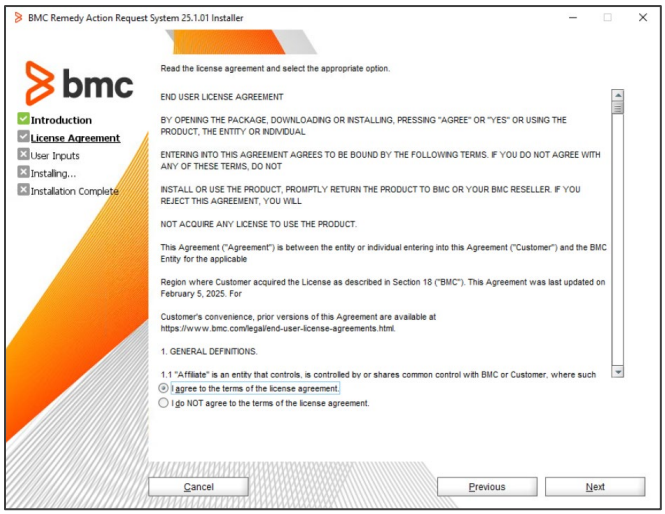



## Task 9: Install AR System Clients

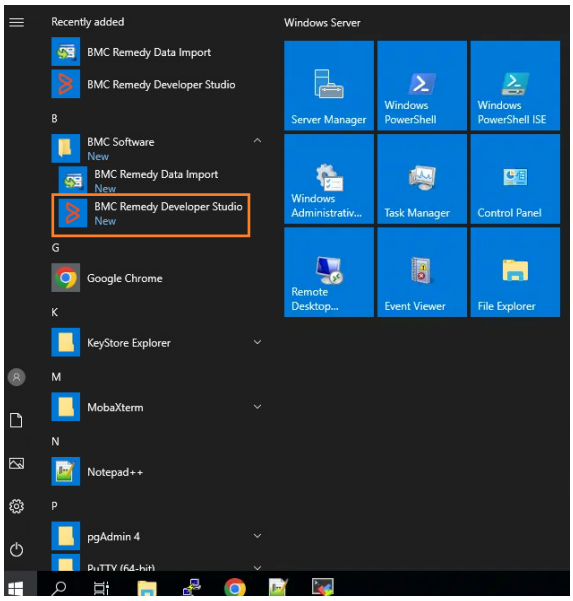
In this task you will install the AR Suite Developers kit for Windows.

1. Using **File Explorer**, **create** a new folder **C:\AR Tools**
2. Go to **ReadyTech\Inbox** folder and copy **ARSuiteKitDeveloperWindows2x.x.xx.zip** to the **C:\AR Tools** folder.
3. In the **C:\AR Tools** folder, extract the **ARSuiteKitDeveloperWindows2x.x.xx.zip** file.
4. In the **Disk 1 folder**, run **setup.exe**.
5. You must **agree to the terms of the license agreement** and then complete the installation wizard by accepting the defaults.

   **IMPORTANT:** For the 64-bit JAVA Path field, specify the JRE path **C:\Program Files\Microsoft\jdk-17.0.17.10-hotspot**




6. Launch **BMC Remedy Developer Studio**.



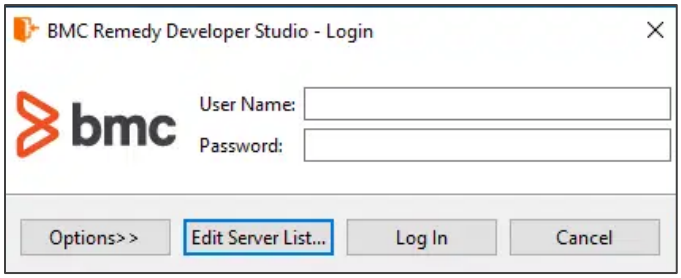

7. In the **BMC Remedy Developer Studio Launcher** dialog box, click **Launch**.
8. To log into **BMC Remedy Developer studio** as hannah_admin user, you need to get the external IP address of the service. Using **MobaXterm**, retrieve the IP address by running this command:

```
kubectl -n helixis get svc | grep platform-admin-ext
```

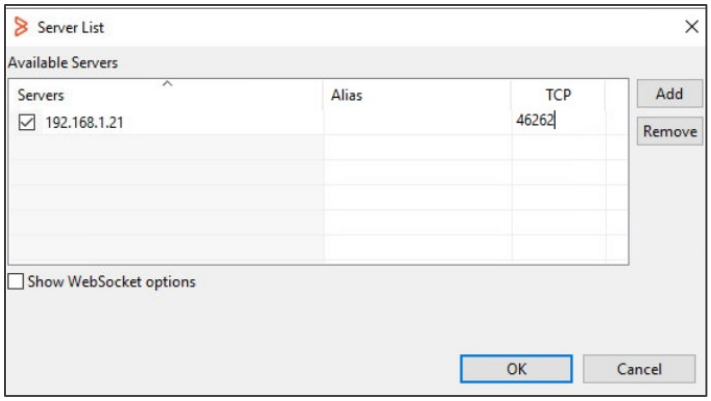
**Note:** This command searches for and displays details about the platform-admin-ext service within the helixis namespace in your Kubernetes cluster.
The output typically includes the type, port, and **two IP addresses**:

- **Cluster IP**: the internal address used for communication within the Kubernetes cluster.
- **External IP**: the public or external-facing address that allows access to the service from outside the cluster (for example, from a web browser or external client).

9. In the **BMC Remedy Developer Studio Login** dialog and click **Edit Server List** to define the server and port to connect.



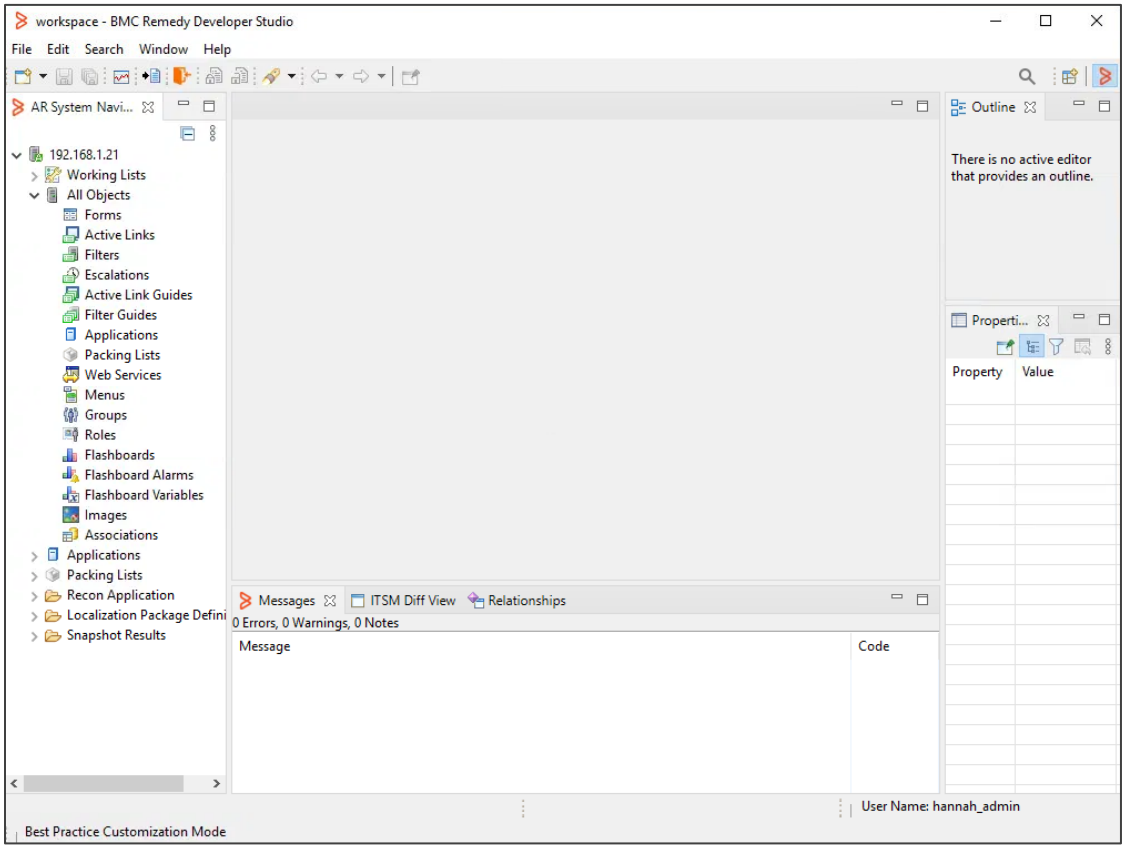

10. Enter the **External Server IP** and **TCP** details gathered in step 7.




11. Login to the server with the following credentials:
    - Username: **hannah_admin**
    - Password: **Passw0rd**!

12. Once logged into the server, expand the **IP Address** (from the left-hand navigation).

13. Double click **All Objects > Forms** link to retrieve the list of AR Forms from the server.



## Task 10: Verify Platform and Application Components

In this task you will perform post-installation checks and verify successful deployment.

1. Verify the platform and application components pod are running by using the following command:

```
kubectl get po -n helixis
```

```
NAME                                                  READY   STATUS      RESTARTS   AGE
atriumwebsvc-84666bc55c-lvnjh                         1/1     Running     0          5d18h
catalog-itsm-plugin-6f946b998-n98g2                   1/1     Running     0          5d16h
catalog-itsm-plugin-6f946b998-vhrk6                   1/1     Running     0          5d16h
dwp-tomcat-deployment-master-84876d8cc4-h4k6w         1/1     Running     0          5d17h
midtier-int-86d9dbb9c4-9tcmq                          1/1     Running     0          5d19h
midtier-user-674c9fbbbf-6flpt                         1/1     Running     0          5d19h
midtiercachebuildercronjob-1639011600-rd69k           0/1     Completed   0          8h
openfire-0                                            1/1     Running     0          5d18h
platform-fts-0                                        1/1     Running     0          5d19h
smartit-master-5db6898f99-cpqvk                       1/1     Running     0          5d18h
smartit-slave-78c8c7f947-4lfg4                        1/1     Running     0          5d18h
smartreporting-5dffbd64d-mqw8b                        1/1     Running     0          5d18h
virtualchatplugin-7c6576d69d-fd6zw                    1/1     Running     0          5d17h
virtualchatserver-0                                   1/1     Running     0          5d17h
```

2. To launch the K9s tool into the helixis namespace, type the following:

```
k9s -n helixis
```

3. On the deployment engine, use the following command to retrieve a list of URLs to access the platform and application components:

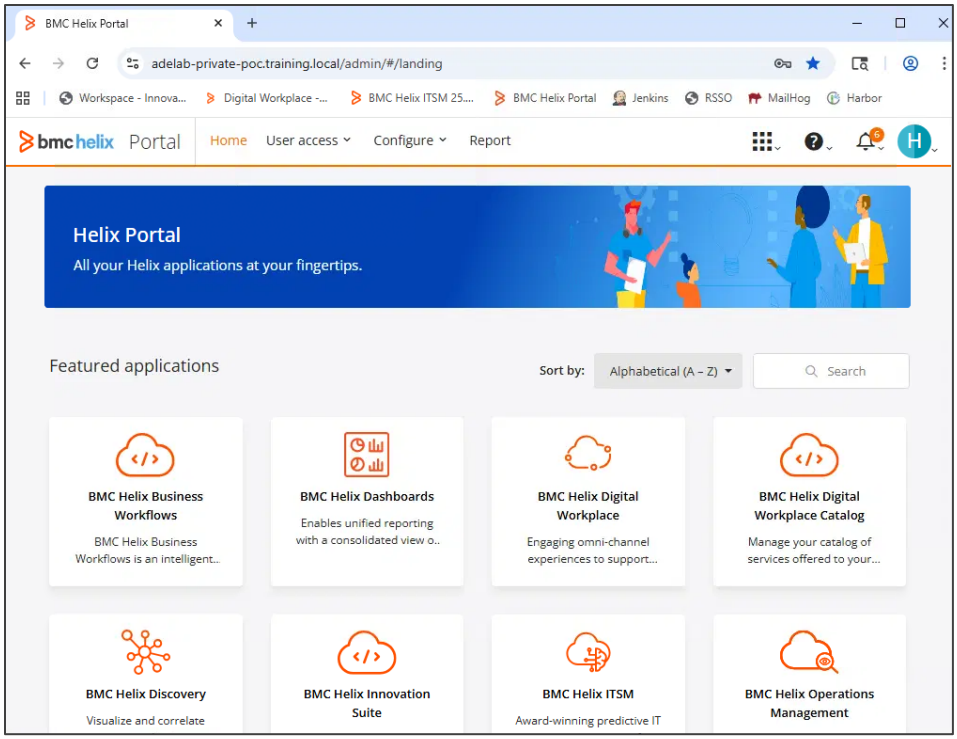
```
kubectl get ing -n helixis
```

```
[root@deploymentengine ~]# kubectl get ing -n helixis
NAME                  CLASS    HOSTS                                          ADDRESS   PORTS   AGE
assisttool-ingress    <none>   is-poc-supportassisttool.training.local                  80      3d13h
atriumwebsvc          <none>   is-poc-atws.training.local                                80      3d17h
catalog               <none>   is-poc-dwpcatalog.training.local                          80      3d13h
dwp-default-ingress   <none>   is-poc-dwp.training.local                                 80      3d13h
midtier-int           <none>   is-poc-int.training.local                                 80      3d17h
midtier-user          <none>   is-poc.training.local                                    80      3d17h
openfire              <none>   is-poc-chat.training.local                               80      3d16h
platform-int          <none>   is-poc-restapi.training.local                            80      3d17h
platform-user         <none>   is-poc-is.training.local                                 80      3d17h
smartit               <none>   is-poc-smartit.training.local                            80      3d16h
virtualchatserver     <none>   is-poc-vchat.training.local                              80      3d16h
```

4. In **Chrome**, log into the Mid Tier:
   - **https://is-poc.training.local/arsys**
   - Username: **hannah_admin**
   - Password: **Passw0rd!**

   **Note:** This password was set in the **HELIX_ONPREM_DEPLOYMENT** pipeline in the **BAKEDUSER_HANNAH_ADMIN_PASSWORD** parameter.

5. In **Chrome**, log into the Helix Portal:
   - **https://adelab-private-poc.training.local**
   - Username: **hannah_admin**
   - Password: **Passw0rd!**




6. Click the following tiles to verify the consoles open successfully:
    - BMC Helix Digital Workplace
    - BMC Helix ITSM

## Task 11: Apply AR System Server license

In this task you will apply application licenses in the Mid Tier.

1. Open the **Bootcamp** page on the **BMC Communities** site (ITSM & ITOM On-Prem Boot Camp Class).

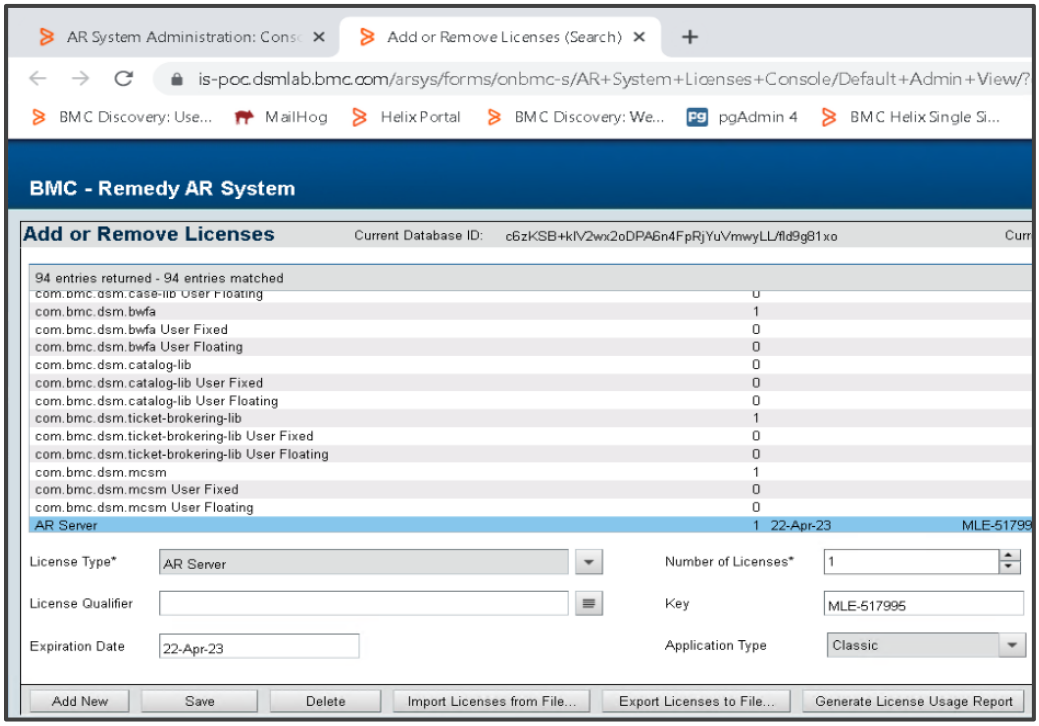
2. Locate this week's **Welcome Message**.

3. From the **Notes & Attachments** section, locate the file titled **Current AR Server License** for the valid AR license. This information will be needed later in the lab.

4. In **Chrome**, log into the Mid Tier:
    - **https://is-poc.training.local/arsys**
    - Username: **hannah_admin**
    - Password: **Passw0rd!**

5. Navigate to **Applications > AR System Administration > AR System Administration Console** to open the AR System Administration Console.

6. Then navigate to **System > General > Add or Remove License**.




7. Click the **Add New** button. Enter the **AR Server** License Type values based upon the information retrieved from the Bootcamp Communities space.

   **Note:** The values in the screenshot may not reflect the information you enter in the lab environment.

8. To validate that the license was added properly, navigate to the **Server Information > Licenses tab**. If the Server License Type indicates "**AR Server**" then the license has been properly applied.

**REQUIRED LAB COMPLETION CHECKPOINT:** Please post a full screenshot of the following pages to the **Bootcamp page** on the **BMC Communities** site to confirm you have successfully completed **Lab 6**. Facilitators may request to connect to your system for additional verification if needed.

- MidTier home page
- Helix Portal page
- Helix ITSM Dashboard

**CONGRATULATIONS!** You have completed the required lab portion of the BMC Helix Technical Bootcamp - ITSM/ITOM Container deployment.